

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Phase 1b/2 Study of BMS-813160 in Combination With Chemotherapy or Nivolumab in Patients With Advanced Solid Tumors **Short Title/Acronym:** A Study of BMS-813160 in Combination With Chemotherapy or Nivolumab in Participants With Advanced Solid Tumors **Protocol Number:** CV202-103 **NCT Number:** NCT03184870 **Sponsor/Company Name:** Bristol Myers Squibb (BMS) **Investigational Product:** BMS-813160 (CCR2/5 dual antagonist), Nivolumab, Nab-paclitaxel, Gemcitabine, fluorouracil / 5-fluorouracil (5-FU), Leucovorin, Irinotecan (FOLFIRI) **Phase of Development:** PHASE1, PHASE2 **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety profile (including Adverse Events and Serious Adverse Events), tolerability, drug levels, drug effects, and pharmacodynamics (incidence of Dose-Limiting Toxicities) of BMS-813160 in combination with chemotherapy with or without nivolumab. To evaluate preliminary efficacy, primarily the Overall Response Rate (ORR) as assessed by Blinded Independent Central Review (BICR) per RECIST v1.1.

Secondary Objectives: To assess pharmacokinetic (PK) parameters, median Duration of Response (DOR), Progression-Free Survival (PFS) rate at 24 weeks, safety profile, and tumor pharmacodynamic parameters (including Tregs, tumor-associated macrophages [TAMs], and tumor-infiltrating lymphocytes).

2.2 Background & Rationale Cysteine-cysteine chemokine receptors 2 (CCR2) and 5 (CCR5) contribute to immune suppression in tumor microenvironments. CCR2 and CCR5 antagonists have demonstrated antitumor activity in pancreatic ductal adenocarcinoma (PDAC) and colorectal cancer (CRC). This study is conducted to address whether blocking this axis with a dual CCR2/5 antagonist (BMS-813160) can overcome the immunosuppressive tumor microenvironment and promote effective anti-tumor immunity when combined with chemotherapy and/or the PD-1 inhibitor Nivolumab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Parallel Assignment **Blinding:** Open-label

(with Blinded Independent Central Review for scans)

Randomization: Non-randomized

3.2 Planned Enrollment & Duration Trial Status: COMPLETED (Study has been completed) **Targeted Enrollment:** 332 participants **Number of Sites:** Multicenter (~40 participating locations) **Estimated Study Start:** 08-Aug-2017 **Estimated Primary Completion:** 06-Apr-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Males and females, age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of metastatic colorectal or pancreatic cancer measurable by RECIST v1.1. 3. Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 1 . 4. Adequate marrow function (e.g., WBC $\geq 2000/\mu\text{L}$, Platelets $\geq 100 \times 10^3/\mu\text{L}$) and adequate other organ functions. 5. Required to undergo mandatory pre- and on-treatment biopsies. 6. Ability to swallow pills or capsules. 7. Ability to comply with study visits, treatment, procedures, pharmacokinetic (PK) and pharmacodynamic (PD) sample collection, and required study follow-up.

4.2 Exclusion Criteria 1. Histology other than adenocarcinoma (e.g., neuroendocrine or acinar cell). 2. Suspected, known, or progressive central nervous system (CNS) metastases (Imaging required only if participants are symptomatic). 3. Active, known, or suspected autoimmune disease. 4. Condition requiring systemic treatment with either corticosteroids (> 10 mg daily prednisone equivalents) or other immunosuppressive medications within 14 days of study treatment administration. 5. Prior treatment with cysteine-cysteine chemokine receptor 2 (CCR2) and/or cysteine-cysteine chemokine receptor 5 (CCR5) inhibitors, programmed death-1 receptor (PD-1), programmed death-ligand 1 [PD(L)-1] or cytotoxic T-lymphocyte antigen-4 (CTLA-4) antibodies. 6. Interstitial lung disease that is symptomatic or may interfere with the detection or management of suspected treatment-related pulmonary toxicity. 7. History of allergy to study treatments or any of its components of the study arm that participant is enrolling.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patients receive a 2-week monotherapy lead-in of BMS-813160 (300 mg BID, 600 mg QD, 300 mg QD, or 150 mg QD). Following lead-in, patients receive BMS-813160 combined with chemotherapy (Gemcitabine/Nab-paclitaxel for PDAC, or FOLFIRI for CRC) $\pm$ Nivolumab. **Investigational Products & Trade Names:** BMS-813160, Nivolumab, Nab-paclitaxel, Gemcitabine, Leucovorin, Irinotecan, and fluorouracil (Trade names: Carac, Efudex, Adrucil, Fluoroplex, Tolak). **Route of**

Administration: Oral (BMS-813160 capsules/tablets) and Intravenous (Nivolumab and Chemotherapy). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard antiemetics, hydration, and palliative supportive care medications allowed for oncology patients. **Prohibited:** Systemic immunosuppressants, concurrent experimental therapies, and corticosteroids exceeding 10 mg/day prednisone equivalent.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans (CT/MRI), Biopsies
Baseline	Day 0	Enrollment, First dose of BMS-813160 monotherapy lead-in
Interim	Every 8 Weeks	Combination Therapy, Imaging via CT/MRI for tumor response, PK/PD sample collection, Safety Labs
End of Study	Varies	Treatment discontinuation, final safety assessment, 125-day safety follow-up (for Nivolumab groups)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Number of Participants Experiencing Adverse Events (AEs):** Defined as any new untoward medical occurrence or worsening of a pre-existing medical condition.
- Number of Participants Experiencing Serious Adverse Events (SAEs):** Defined as any untoward medical occurrence that results in death, is life-threatening, requires hospitalization, or prolongs existing hospitalization.
- Number of Participants Experiencing Dose-Limiting Toxicities (DLTs):** Severe AEs attributed to the regimen used to define the maximum tolerated dose. Evaluated during the first 6 weeks of treatment in Part 1 and 4 weeks for Part 2.
- Overall Response Rate (ORR):** Assessed by Blinded Independent Central Review (BICR) per RECIST v1.1.

7.2 Secondary & Exploratory Endpoints

- * Maximum Concentration (C_{max})
- * Time to Maximum Concentration (T_{max})
- * Trough Observed Plasma Concentration (C_{trough})
- * Median Duration of Response (DOR).
- * Progression-Free Survival (PFS) rate at 24 weeks.
- * Changes in tumor pharmacodynamic parameters (Tregs, TAMs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical exams, laboratory tests, and continuous patient monitoring. Graded on a scale from 1 (mild) to 5 (fatal) per NCI CTCAE v4.03.

Serious Adverse Events (SAEs): Evaluated strictly through 125 days post-last administration of immunotherapy regimens. Dose-

limiting toxicities reported urgently. **Data Monitoring Committee (DMC):** Continual toxicity monitoring managed by Bristol Myers Squibb to evaluate exceeding unacceptable DLT rates.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy assessed in all patients who received at least one dose of study drug and had measurable baseline disease with ≥ 1 post-baseline scan. Safety set included all treated patients. **Sample Size Justification:** A Simon optimal 2-stage design was utilized to test the null hypothesis regarding the true disease control rate and ORR compared to historical benchmarks (e.g., Gem/Abraxane MPACT study). **Handling of Missing Data:** Participants dropping out prior to first evaluable scan without documented clinical progression were typically treated as non-responders in ORR calculations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite data verification, central lab reviews of PK/PD, and independent central review for RECIST endpoints. **Audit Trail:** Standard tracking of eCRF modifications, dose modifications, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CV202-103 **Registry Number:** NCT03184870 **Posting ID:** 22720114948792005 **Sponsor/Lead Organization:** Bristol Myers Squibb (BMS) **Study Title:** A Study of BMS-813160 in Combination With Chemotherapy or Nivolumab in Participants With Advanced Solid Tumors **Official Regulatory Title:** A Phase 1b/2 Study of BMS-813160 in Combination With Chemotherapy or Nivolumab in Patients With Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Disease Areas & Primary Conditions: Gastrointestinal Neoplasms, specifically Metastatic Pancreatic Cancer (Pancreatic Ductal Adenocarcinoma - PDAC), Colorectal Cancer, and Rectal Cancer (Preferred UMLS Name: Rectal Carcinoma). **Diagnosis Threshold:** Histologically/cytologically confirmed measurable metastatic disease **Medical Methodology:** Dual CCR2/5 antagonism combined with PD-1 blockade and/or standard cytotoxic chemotherapy **Optimization Target:** Tumor volume reduction (ORR) and sustained disease control

13. Study Procedures & Methodology

Management Pathway: Standard oncological RECIST v1.1 evaluation pathway
Education Protocol (HCP): Site initiation visits covering the novel dual antagonist toxicity profile and trial-specific biopsy protocols
Education Protocol (Patient): Informed consent, biopsy expectations, and reporting mechanisms for immune-related adverse events
Quality Audit Mechanism: Blinded Independent Central Review (BICR) of all imaging and central lab validation of biopsies

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity
Onsite Visit Cadence: Every 8 weeks for primary imaging assessments
Remote Monitoring Frequency: Intermittent follow-ups for survival/progression post-treatment
Checklist Review Interval: Prior to the initiation of every new treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				